

Table 1: Theorem 1 empirical check — SMAC (SafeDreamer, dead_allies.incremental cost). Does the bound Δ cover the actual gap between real and imagined cost?

Env	d	lag-lr	ϵ_c	TV (from KL)	Δ (bound)	Observed gap	Holds?	Slack	Seeds
1c3s5z	0.0	1e-05	0.0156 ± 0.0013	0.3684 ± 0.0041	38.9 ± 0.4	0.3138 ± 0.1669	✓	$\times 124$	3
2m _v s1z	0.0	1e-05	0.0008 ± 0.0006	0.1456 ± 0.0022	15.3 ± 0.2	0.1913 ± 0.1831	✓	$\times 80$	3
2s3z	0.0	1e-05	0.0099 ± 0.0001	0.2773 ± 0.0019	29.3 ± 0.2	0.1961 ± 0.1151	✓	$\times 149$	3
2s _v s1sc	0.0	1e-05	0.0009 ± 0.0001	0.1532 ± 0.0087	16.1 ± 0.9	0.0250 ± 0.0170	✓	$\times 644$	3
3m	0.0	1e-05	0.0021 ± 0.0003	0.1699 ± 0.0042	17.9 ± 0.4	0.0401 ± 0.0194	✓	$\times 446$	3
3s _v s3z	0.0	1e-05	0.0016 ± 0.0002	0.2494 ± 0.0109	26.2 ± 1.1	0.0279 ± 0.0240	✓	$\times 939$	3
3s _v s4z	0.0	1e-05	0.0005 ± 0.0000	0.2271 ± 0.0097	23.9 ± 1.0	0.0732 ± 0.1000	✓	$\times 326$	3
3s _v s5z	0.0	1e-05	0.0007 ± 0.0006	0.2370 ± 0.0175	24.9 ± 1.8	0.1835 ± 0.3327	✓	$\times 136$	6
8m	0.0	1e-05	0.0070 ± 0.0004	0.2565 ± 0.0052	27.0 ± 0.5	0.0846 ± 0.0204	✓	$\times 319$	3
MMM	0.0	1e-05	0.0155 ± 0.0052	0.2839 ± 0.0178	30.0 ± 1.8	0.5160 ± 0.7064	✓	$\times 58$	6

Formula:

$\Delta = 15\epsilon_c + 105 \cdot c_{\max} \cdot \text{TV}$, $c_{\max} = 1$ (binary dead_allies.incremental cost).

What each column measures:

- ϵ_c — *Cost-head error*. How wrong the model prices each step’s cost. Source: average of `Model/cost_loss` over the last 10% of training rows.
- **TV** — *Transition error*. How wrong the model’s next-state forecast is. Source: average of `Model/div` (KL divergence, prior vs. posterior) over the last 10% of training rows. Converted to TV via Pinsker’s inequality: $\text{TV} = \sqrt{\text{KL}/2}$.
- **Δ (bound)** — The maximum gap Theorem 1 *allows* between real and imagined cost.
- **Observed gap** — The gap we *actually measured*: $|J_{\text{real}} - \hat{J}_{\text{imag}}|$.
Both sides scaled to a discounted 15-step window ($\times 13.9 = \sum_{t=0}^{14} \gamma^t$, $\gamma = 0.99$):
— Imagined: `Value/Cost` (per dreamed step) $\times 13.9$.
— Real: `main/cost` (episode total) $\div$ episode length $\times 13.9$.
Episode length = measured from consecutive `steps` diffs (not hardcoded).
- **Slack** — How much room to spare: $\Delta \div \text{gap}$.

Assumptions & caveats:

- **Average, not worst-case.** Theorem 1 requires uniform bounds on every state-action pair. All values here are training averages. This table is a consistency check (“does the data contradict the theorem?”), not a worst-case proof.
- **Tail-10% window.** We average each metric over the last 10% of logged rows, assuming the model has converged by that point.
- **Even cost spread.** Real per-step cost = episode total $\div$ episode length. This assumes cost is spread evenly, but cost is binary (0 or 1 per step) and likely bursty — risky windows may have much higher cost than the average suggests.
- **KL $\rightarrow$ TV conversion.** Pinsker gives a conservative upper bound on TV from KL, but both are computed on latent codes, not raw states. The theorem is stated over the true state space.